LegalEase

AI-Powered Legal Document Generation

PROJECT REPORT

Project Type	Academic Software Project
Technology Focus	Artificial Intelligence • Natural Language Processing • Web Application
Target Users	Students, individuals, small businesses and legal-support users
Prepared By	_____
Register No.	_____
Department	B.Sc. Computer Science
Academic Year	2026–2027

A practical prototype that uses AI-assisted natural language processing to collect user requirements and generate structured legal-document drafts.

CERTIFICATE

This is to certify that the project report entitled “LegalEase: AI-Powered Legal Document Generation” is a bonafide work carried out as part of the academic project requirements for the Bachelor of Science in Computer Science. The work demonstrates the design of an AI-assisted application for generating structured legal-document drafts from user-provided information.

_____	_____	_____
Project Guide	Head of the Department	External Examiner

DECLARATION

I hereby declare that the project titled “LegalEase: AI-Powered Legal Document Generation” is prepared for academic purposes and represents my understanding of artificial intelligence, natural language processing, web application development, document generation and responsible software design. The prototype is intended as an educational demonstration and does not replace professional legal advice.

ACKNOWLEDGEMENT

I express my sincere gratitude to my project guide, department faculty and institution for their guidance and support. I also acknowledge the learning resources and open technologies that helped in understanding AI, NLP, web development and document automation. This project helped me connect theoretical computer-science concepts with a practical application.

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1. ABSTRACT

LegalEase is an AI-assisted web application concept designed to simplify the preparation of common legal-document drafts. Instead of starting with a blank document, the user selects a document type and answers guided questions. The system validates basic inputs, converts the information into structured fields, and uses an AI/NLP layer to create a readable draft. The final document can be reviewed and exported as a document or PDF.

The project focuses on automation, usability and responsible AI. It is designed as an academic prototype rather than a system that independently provides legal advice or guarantees legal validity. A human review step is therefore included before final use.

- Reduce repetitive manual formatting and drafting effort.
- Provide guided data collection instead of requiring users to know legal document structure.
- Generate consistent sections, placeholders and formatting.
- Maintain an audit trail of generated drafts and revisions.
- Demonstrate practical use of AI/NLP in a real-world workflow.

2. INTRODUCTION

Legal documents often contain repeated structures such as headings, parties, dates, addresses, clauses, declarations and signatures. For a student project, this makes document automation a suitable application area for AI and software engineering.

LegalEase combines a simple web interface, structured form processing, template management and an AI generation layer. The user remains responsible for checking the generated content and obtaining professional advice where appropriate.

3. PROBLEM STATEMENT

Creating a formal document from scratch can be time-consuming for users who are unfamiliar with document structure. Manual copying can also introduce missing fields, inconsistent formatting or outdated wording. The project addresses the software problem of converting user-provided facts into a structured, editable legal-document draft while preserving user control.

4. OBJECTIVES

- Create a simple interface for selecting and generating document drafts.
- Collect information through structured forms and guided prompts.
- Use NLP/AI to transform structured information into natural-language sections.
- Apply templates and validation rules to improve consistency.
- Allow preview, editing, versioning and export.
- Protect sensitive information through access control and secure storage practices.
- Clearly communicate that generated content requires human review.

5. EXISTING SYSTEM

Traditional workflows commonly rely on blank word-processing files, reusable templates or manual assistance. These approaches can work well for experienced users, but they may require repeated formatting and careful checking.

- Manual entry and formatting take additional time.
- Users may not know which information is required.
- Copy-paste workflows can create inconsistent documents.
- Changes to a template may not be reflected in older copies.
- There is limited automation between user input and final document generation.

6. PROPOSED SYSTEM

The proposed LegalEase system introduces a guided workflow: Document Type → User Details → Validation → AI/NLP Generation → Template Formatting → Preview → Human Review → Export.

The AI layer is used to assist with language generation and organization. Deterministic rules handle required fields, formatting and validation wherever possible. This separation makes the prototype easier to test and reduces unnecessary dependence on generated text.

7. SYSTEM REQUIREMENTS

Category	Requirement
Hardware	4 GB RAM or above; dual-core processor or better; 10 GB free storage.
Client	Modern browser such as Chrome, Edge or Firefox.
Frontend	HTML5, CSS3, JavaScript; optional React for a richer interface.
Backend	Python Flask/FastAPI or Node.js/Express.
AI/NLP	LLM/API or locally hosted language model; prompt/template layer.
Database	SQLite for prototype; PostgreSQL/MySQL for deployment.
Document Engine	Python-docx / ReportLab or equivalent document-generation library.

8. SYSTEM ARCHITECTURE

USER	WEB UI	APPLICATION SERVER	AI / NLP	DOCUMENT STORE
Select type Enter facts Review draft	Forms Validation Preview	Auth Business rules Template engine	Prompting Generation Consistency checks	Drafts Versions Export files

Data Flow

- 1. User chooses a supported document category.
- 2. The interface displays only the fields relevant to that category.
- 3. Backend validation checks mandatory values and data formats.
- 4. Structured data is combined with a controlled template/prompt.
- 5. AI/NLP produces the draft sections.
- 6. Post-generation checks identify missing fields or unsupported output.
- 7. User reviews and edits the draft.
- 8. The system stores the approved version and generates an export file.

9. MODULES

- User Management: registration/login, role control and session management.
- Document Template Manager: stores document types, required fields and clause templates.
- Input & Validation: collects user data and validates dates, names, contact fields and required values.
- AI/NLP Engine: generates or rewrites sections according to controlled instructions.
- Document Generator: inserts content into a consistent layout and produces PDF/DOCX output.
- Review & Versioning: lets the user edit the generated draft and maintain versions.
- Audit Module: records creation time, document type and generation status without unnecessarily exposing sensitive text.

10. AI / NLP WORKFLOW

The project can use an AI language model through an API or a local model. A controlled prompt is created from the selected document type and structured user fields. The system should instruct the model to avoid inventing facts and to preserve supplied names, dates and amounts.

A useful design is structured input + template constraints + AI drafting + deterministic validation. This is more reliable than asking the model to create an entire legal document from an unrestricted natural-language request.

- Prompt builder converts form data into a consistent request.
- Generation layer produces section text.
- Validation layer checks required placeholders and prohibited missing information.
- Human review remains mandatory before final use.

11. DATABASE DESIGN

Table	Important Fields	Purpose
users	user_id, name, email, password_hash, role	Authentication and roles
document_types	type_id, name, description, version	Supported document categories
templates	template_id, type_id, title, content, version	Reusable structures and clauses
documents	document_id, user_id, type_id, status, created_at	Document records
document_data	document_id, field_name, field_value	Structured user inputs
versions	version_id, document_id, content, created_at	Draft history
audit_logs	log_id, user_id, action, timestamp	Activity tracking

12. ALGORITHM / PSEUDOCODE

```
START
1. Display supported document types.
2. Read selected document type.
3. Load required fields and template rules.
4. Collect user information.
5. Validate required fields and formats.
6. Build structured prompt/context from validated fields.
7. Generate draft sections using AI/NLP.
8. Check output for missing placeholders and unsupported claims.
9. Display editable preview.
10. Save approved version.
11. Generate PDF/DOCX export.
END
```

13. USER INTERFACE DESIGN

- Dashboard: New Document, My Documents, Templates and Profile.
- Document Selection: cards for supported categories.
- Smart Form: simple questions with explanations and required-field indicators.
- Draft Preview: side-by-side form data and generated document preview.
- Review Screen: edit, regenerate a section, save version and export.
- History: previous versions with date and status.

14. IMPLEMENTATION PLAN

- Phase 1 – Requirement analysis and document-type selection.
- Phase 2 – UI design and database schema.
- Phase 3 – Backend APIs and authentication.
- Phase 4 – Template engine and document generation.
- Phase 5 – AI/NLP integration and prompt design.
- Phase 6 – Testing, security checks and usability improvements.

- Phase 7 – Final documentation and demonstration.

15. TESTING

Test Case	Input / Action	Expected Result
TC01	Open application	Dashboard loads correctly
TC02	Leave required field empty	Validation message is displayed
TC03	Enter valid user details	Data is accepted
TC04	Generate draft	Structured draft is created
TC05	Edit generated section	Edited text is saved
TC06	Create new version	Previous version remains available
TC07	Export PDF	Readable PDF is generated
TC08	Unauthorized document access	Access is denied

16. SECURITY, PRIVACY & RESPONSIBLE USE

Legal information can be sensitive. The prototype should minimize collection, protect stored data and make access decisions on the server. Passwords should be stored as secure hashes rather than plain text. Production deployment should use HTTPS and appropriate database access controls.

AI-generated text can contain omissions or incorrect statements. LegalEase therefore treats AI output as a draft. The interface should display a clear review notice and should never imply that generated text is automatically legally valid for every jurisdiction or situation.

- Use role-based authorization for documents.
- Encrypt sensitive data at rest where appropriate.
- Avoid logging full document contents unless required.
- Provide deletion and retention controls.
- Validate file uploads and generated files.
- Keep templates versioned so changes are traceable.

17. ADVANTAGES & LIMITATIONS

Advantages

- Faster preparation of structured drafts.
- Beginner-friendly guided workflow.
- Consistent formatting through templates.
- Editable drafts and version history.
- Good demonstration of AI + web development + database integration.

Limitations

- AI output may contain errors or ambiguous wording.
- Legal requirements vary by jurisdiction and document type.
- The prototype may support only a limited set of document categories.
- High-quality deployment requires stronger security, legal review and testing than an academic prototype.

18. FUTURE ENHANCEMENTS

- Multilingual document drafting, including Indian regional languages.
- Voice-based data collection for accessibility.
- Digital-signature integration.
- Document comparison and clause-change highlighting.
- Jurisdiction-aware template libraries maintained by qualified professionals.
- Retrieval-augmented generation using approved legal sources.
- Stronger automated consistency checks for names, dates, amounts and references.
- Cloud deployment with encrypted storage and organization-level access control.

19. CONCLUSION

LegalEase demonstrates how artificial intelligence can be combined with conventional software engineering to automate a document-heavy workflow. The key design principle is to use AI as an assistant within a controlled pipeline rather than treating generated text as an unquestionable final answer.

As an academic project, the system provides practical exposure to UI design, backend development, databases, NLP, prompt engineering, document generation, testing and responsible AI. With expert legal review and stronger production controls, the concept can be extended into a broader document-assistance platform.

20. REFERENCES

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6. Python-docx Documentation – Programmatic DOCX generation.

7. General academic literature on Natural Language Processing, Large Language Models and Responsible AI.

PROJECT END